The Mechanism of Fields: Geometric Extension and Optical Examples

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This supplement to *The Mechanism of Fields from the Bost–Connes System* calculates the consequences of specified common time and length scales and a nonlinear transport relation. These relations have not yet been derived from the original complete process adopted in the main text; the optical coefficients below belong to this extension model, while the actual spatial propagation coefficient in the source background still requires independent determination. The original field model [1] provides the form of the extension, and this supplement retains its reproducible results and their scope of applicability.

Contents

S1 Common Time and Length Scales and the Exponential Distribution **3**
S1.1 Geometric Relations 3
S1.2 Conserved Flow and the Exponential Solution 3
S1.3 Optical Path and Time Delay 4

S2 Source Coupling and Motion Response **5**
S2.1 Comparison of Two Couplings 5
S2.2 Free Flow and a Complete Source at Rest 5

S3 Optical Structure of the Exponential Geometry **7**
S3.1 Weak-Field Benchmark 7
S3.2 Second-Order Light Deflection 7
S3.3 Strong-Field Light Paths 8

References **11**

§S1 Common Time and Length Scales and the Exponential Distribution

§S1.1 Geometric Relations

Let A denote the frequency ratio of a local clock to the distant baseline, and B the ratio of local proper length to reference coordinate length. The common isotropic geometry and its mass-shell relation are written as [2]

$$ds^2 = A^2 dt^2 - B^2 d\boldsymbol{x}^2, \qquad E = A\sqrt{m^2 + \frac{|\boldsymbol{p}|^2}{B^2}}. \tag{S1}$$

This example specifies $B = A^{-1}$, giving the spatial propagation coefficient $A/B = A^2$. This also specifies how matter and light propagation change with the clock rate; the rest phase mA alone cannot determine this relation. In the weak field, let $A = 1 + \Phi$. Then

$$\boxed{ds^2 = (1 + 2\Phi)dt^2 - (1 - 2\Phi)d\boldsymbol{x}^2 + O(\Phi^2).} \tag{S2}$$

The corresponding optical refractive index is $n_{\text{opt}} = B/A = A^{-2}$. Light paths and arrival times are determined by this spatial distribution.

§S1.2 Conserved Flow and the Exponential Solution

Let n_i denote the background deficit being transported, and $\pi_i > 0$ the stationary weight of the same network. For exchange satisfying detailed balance, let the jump rate from i to j be w_{ij}, and the conductance be $\kappa_{ij} = w_{ij}\pi_i = w_{ji}\pi_j$. The net flow from i to j and the stationary source equation are [3]

$$J_{ij} = \kappa_{ij}\left(\frac{n_i}{\pi_i} - \frac{n_j}{\pi_j}\right), \qquad \sum_j J_{ij} = s_i. \tag{S3}$$

This formulation retains local occupation, stationary weight, and exchange capacity separately. Constant coefficients recover the harmonic distribution of the main text; when the coefficients change with the field, flow remains conserved in the same way, while the spatial profile adjusts accordingly.

The original field model [1] adopts the following nonlinear constitutive relation. Write $g = A^2 > 0$, take the deficit to be $n = n_0(1 - g)$ and the stationary weight to be $\pi = \pi_0 g$, and let the conductance measured per unit common reference time be $\kappa = \kappa_0 g$. Here n_0, π_0, κ_0 are constants. These relations connect the field-induced changes to both deficit measurement and the transport coefficient, specifying a definite extension model.

The continuum representation retains the dimensionless normalization of n/π; J_r denotes the radial net flow per unit common reference time and per unit reference coordinate area, with the corresponding transport coefficient κ absorbing the conversions involving lattice spacing and cell volume. In the spherically symmetric region outside the source, Eq. (S3) becomes

$$J_r = -\kappa\frac{d}{dr}\left(\frac{n}{\pi}\right) = \frac{\kappa_0 n_0}{\pi_0}\frac{d\log g}{dr}, \qquad 4\pi r^2 J_r = Q. \tag{S4}$$

Here Q is the net source flow within the sphere. Calibrating the far-field amplitude by GM_s and imposing $g \to 1$ at large distances, integration gives

$$\log g = -\frac{2GM_s}{r}, \qquad \boxed{A = e^{-GM_s/r}, \quad g_{00} = e^{-2GM_s/r}, \quad B^2 = e^{2GM_s/r}.} \tag{S5}$$

The exponential form follows from the specific constitutive relation in Eq. (S4). It preserves conservation of the steady flow and reduces in the weak field to $A = 1 - GM_s/r + \cdots$, thereby connecting to the leading gravitational potential of the main text. At finite field strength, the phase potential $\Phi = A - 1$ must be distinguished from the logarithmic clock rate $\log A$: the latter is the harmonic quantity outside the source in this model, and the two coincide only at leading order.

This model explicitly assigns departures beyond the weak field to the transport structure. Expanding in $x = GM_s/r$, the temporal and spatial components are $g_{00} = 1 - 2x + 2x^2 + \cdots$ and $B^2 = 1 + 2x + 2x^2 + \cdots$, respectively. Within the domain of applicability of the continuum model, A remains positive for finite positive r; near the source interior, the actual source structure and discrete exchange must also be used. The external exponential solution alone does not determine the interior geometry.

§S1.3 Optical Path and Time Delay

The arrival time of light also reads the spatial length scale. In the geometric-optics representation [4], the null-interval condition gives

$$dt = n_{\mathrm{opt}}(\boldsymbol{x})\, d\ell, \qquad n_{\mathrm{opt}} = \frac{B}{A}, \qquad \mathcal{T}_\gamma = \int_\gamma n_{\mathrm{opt}}\, d\ell. \tag{S6}$$

Here $d\ell = |d\boldsymbol{x}|$, and $\mathcal{T}_\gamma$ is the reference propagation time along the path. The optical refractive index n_{opt} is given by the relation between time and length scales. Light paths change according to this optical length, and the propagation-time difference between two paths can be converted into a phase difference; a round-trip signal connects emission, reflection, and reception into a distance measurement [5].

With $B = A^{-1}$ as specified in this section, the weak-field relation is $n_{\mathrm{opt}} = A^{-2} = 1 - 2\Phi + O(\Phi^2)$. For a weak field outside the source and fixed distant emission and reception positions, the leading delay is obtained along the zeroth-order path:

$$\delta\mathcal{T} = -2\int_\gamma \Phi\, d\ell, \qquad \delta\varphi = E_\infty\, \delta\mathcal{T}. \tag{S7}$$

Here E_∞ is the light frequency relative to the distant reference time. The second expression compares monochromatic propagation phases at fixed E_∞ and a common detection time; if wave-packet arrival times are compared, the first expression is read out directly. A negative gravitational potential increases the propagation time. Time delay and light deflection jointly reflect the spatial variation of B/A, with their magnitudes determined by the relation between common time and length scales specified in this section.

§S2 Source Coupling and Motion Response

§S2.1 Comparison of Two Couplings

The same response for a rest mass can arise from different couplings in motion. At fixed canonical momentum, we compare modulation of the mass term alone with modulation of the common time and length scales in S1. Each corresponds to its own action, and together they do not uniquely determine the source strength of freely propagating light.

The quantity m^2/E in Eq. (4.2) of the main text can correspond to a definite coupling. Keep the displacement term unchanged and change only the mass term according to $m \mapsto (1+\lambda)m$. The positive-energy branch and its first-order response are then

$$E_\lambda = \sqrt{|\boldsymbol{p}|^2 + (1+\lambda)^2 m^2}, \qquad w_m = \left.\frac{\partial E_\lambda}{\partial \lambda}\right|_{\lambda=0} = \frac{m^2}{E}. \tag{S8}$$

Here λ denotes dimensionless mass modulation. It connects the phase quantity along the trajectory to a specific action: if the source reads precisely this mass-modulation operator, weak readout yields the weight w_m. This mass term vanishes for light, so light produces no source for this action and acquires no propagation response from this term.

The extension using common time and length scales in S1 also changes spatial propagation. Substituting $B = A^{-1}$ and $A = 1+\Phi$ into Eq. (S1), we obtain at fixed canonical momentum

$$E_\Phi = A\sqrt{m^2 + A^2|\boldsymbol{p}|^2}, \qquad w_{\mathrm{geom}} = \left.\frac{\partial E_\Phi}{\partial \Phi}\right|_{\Phi=0} = E + \frac{|\boldsymbol{p}|^2}{E}. \tag{S9}$$

At rest, both responses equal m. Although the mass term for light vanishes, the propagation term in the common time and length scales still changes, giving $w_{\mathrm{geom}} = 2E_\gamma$. This is precisely why the same geometric extension can change the path and phase of light.

If E_Φ is further used as the Hamiltonian shared by matter and the dynamical background, the variational source of matter acting on the background is also given by $\partial E_\Phi/\partial\Phi$. The light response above is then retained together with its corresponding back-reaction. For a source procedure that reads only the mass term, the complete back-reaction must be calculated together with the auxiliaries and controls that implement it. The two responses are each paired with the source and back-reaction determined by their respective joint Hamiltonians.

§S2.2 Free Flow and a Complete Source at Rest

In a local continuum representation using the same normalized momentum measure $d\Gamma_p$, the distribution function $f(\boldsymbol{x}, \boldsymbol{p})$ gives

$$\rho_E = \int d\Gamma_p\, f E, \qquad \Pi^{ij} = \int d\Gamma_p\, f \frac{p^i p^j}{E}, \qquad \rho_E - \operatorname{tr}\Pi = \int d\Gamma_p\, f \frac{m^2}{E}. \tag{S10}$$

Mass-term modulation corresponds to the combination with the minus sign; modulation of the common time and length scales in Eq. (S9) corresponds to $\rho_E + \operatorname{tr}\Pi$.

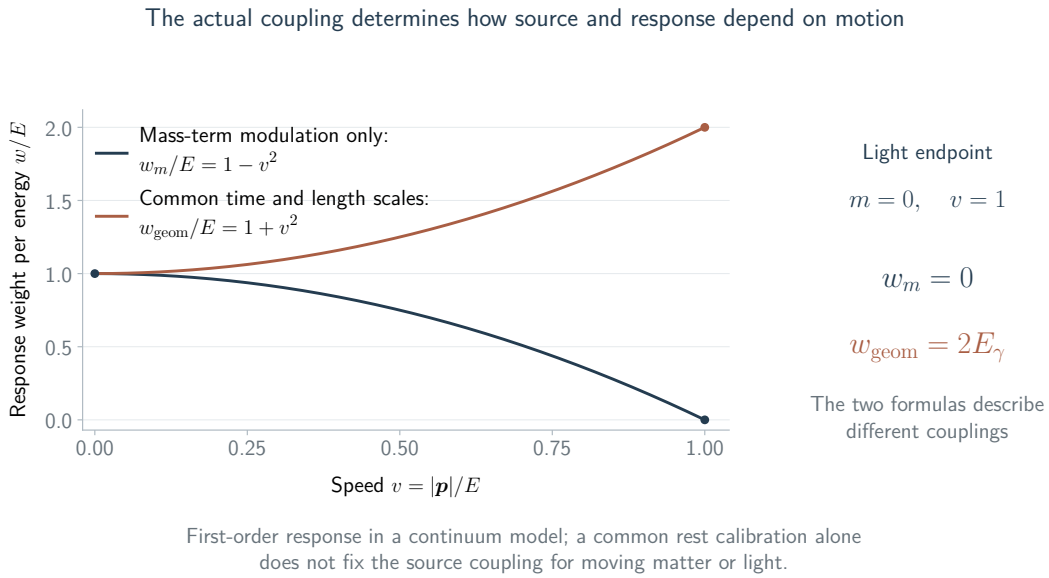


Figure S1. Identical responses at rest and different responses in motion. The two couplings in Eqs. (S8) and (S9) both have weight m at rest, while in motion they give $w_m/E = 1 - v^2$ and $w_{\mathrm{geom}}/E = 1 + v^2$, respectively, where $v = |\boldsymbol{p}|/E$. Light lies at the endpoint $v = 1$.

In the isotropic case, these are $\rho_E - 3P$ and $\rho_E + 3P$, respectively. Freely propagating light satisfies $\mathrm{tr}\,\Pi = \rho_E$, but a complete composite system at rest must also include the stresses of the confining matter and interactions.

In a flat continuum representation neglecting self-gravity, consider a localized system at rest without external support. When its stress–energy tensor is time-independent and surface terms vanish, the complete stress satisfies $\partial_k T^{kj} = 0$. Integrating $\partial_k(x^i T^{kj})$ then gives [6]

$$\int d^3x\, T^{ij} = 0, \qquad \int d^3x \left(T^{00} \pm \sum_i T^{ii} \right) = \int d^3x\, T^{00} = M. \tag{S11}$$

The two complete stress combinations in Eq. (S11) integrate to the total energy for this system at rest. Whether an actual coupling reads either combination is determined by its complete Hamiltonian, including the confining matter and interactions. Agreement for a source at rest therefore cannot by itself select the moving-source coupling; actual writing remains determined by the source operator in Eq. (4.3) of the main text.

§S3 Optical Structure of the Exponential Geometry

§S3.1 Weak-Field Benchmark

Retaining the far-field strength calibration of the main text, write $\mu = GM_s$ and use $c_0 = 1$. This section calculates the spherically symmetric external exponential geometry specified in S1 and compares it with the Schwarzschild geometry having the same far-field mass. Their optical differences test this geometric extension.

The mathematical form of the exponential geometry and its light orbits have been studied independently [7]. In the present example, this geometry is connected to the constitutive relation for background transport in Eqs. (S3)–(S5). For comparison with general relativity, both geometries use the isotropic radius r, with time at large distances normalized to $A(\infty) = 1$. Let $x = \mu/r$. The two static geometries are

$$\begin{aligned} A^2_{\mathrm{exp}} &= e^{-2x}, \\ B^2_{\mathrm{exp}} &= e^{2x}, \\ A^2_{\mathrm{GR}} &= \left(\frac{1 - x/2}{1 + x/2}\right)^2, \\ B^2_{\mathrm{GR}} &= (1 + x/2)^4. \end{aligned} \tag{S12}$$

Here GR denotes the external Schwarzschild solution, whose static exterior coordinate range is $r > \mu/2$. The weak-field expansions give

$$\begin{aligned} A^2_{\mathrm{exp}} &= 1 - 2x + 2x^2 - \frac{4}{3}x^3 + O(x^4), \\ A^2_{\mathrm{GR}} &= 1 - 2x + 2x^2 - \frac{3}{2}x^3 + O(x^4), \\ B^2_{\mathrm{exp}} &= 1 + 2x + 2x^2 + O(x^3), \\ B^2_{\mathrm{GR}} &= 1 + 2x + \frac{3}{2}x^2 + O(x^3). \end{aligned} \tag{S13}$$

The two geometries have the same Newtonian potential, leading clock-rate change, and first-order optical refractive index. Leading free fall, redshift, and light deflection therefore share the same weak-field benchmark. Differences first enter at second order in the spatial component and third order in the temporal component.

These coefficients organize the calculation, while the final comparison concerns actual observables. Light deflection is defined by the distant incoming and outgoing directions; clock-rate comparisons use actual clocks and signals at both ends; and the source mass is determined by the same far-field action. The coordinate radius of closest approach differs from the light ray's impact parameter. Directly substituting the former into a formula for the latter changes the apparent higher-order coefficients.

§S3.2 Second-Order Light Deflection

Light paths are determined by $n_{\mathrm{opt}} = B/A$. Consider a ray incident from a large distance that passes around a spherically symmetric source and returns to a large distance. Define its impact parameter by the ratio of angular momentum to distant

energy, $b = L_\gamma/E_\infty$. In the asymptotically flat region, this is the perpendicular distance between the incoming straight line and the center. Let $u = b/r$ and $\epsilon = \mu/b$, and parameterize the orbit by the polar angle φ in its plane. The light orbit satisfies

$$\left(\frac{du}{d\varphi}\right)^2 = n_{\rm opt}^2(\epsilon u) - u^2, \qquad u(0) = 0, \quad u'(0) = 1. \tag{S14}$$

Here $n_{\rm opt}(x)$ is expressed in terms of the dimensionless position variable x. The exponential model gives $n_{\rm opt}^2 = e^{4\epsilon u}$, hence

$$u'' + u = 2\epsilon e^{4\epsilon u} = 2\epsilon + 8\epsilon^2 u + O(\epsilon^3). \tag{S15}$$

Solving order by order with the incoming conditions gives

$$u(\varphi) = \sin\varphi + 2\epsilon(1 - \cos\varphi) + 4\epsilon^2(\sin\varphi - \varphi\cos\varphi) + O(\epsilon^3). \tag{S16}$$

The outgoing direction is the next point at which $u = 0$. Writing it as $\varphi = \pi + \alpha$ gives the deflection angle. The Schwarzschild geometry uses $n_{\rm GR}^2 = 1 + 4x + 15x^2/2 + \cdots$, yielding the corresponding result under the same incoming conditions and at the same impact parameter:

$$\boxed{\begin{aligned} \alpha_{\rm exp} &= 4\epsilon + 4\pi\epsilon^2 + O(\epsilon^3), \\ \alpha_{\rm GR} &= 4\epsilon + \frac{15\pi}{4}\epsilon^2 + O(\epsilon^3). \end{aligned}} \tag{S17}$$

The second-order difference is $\Delta\alpha = \pi\epsilon^2/4 + O(\epsilon^3)$. Relative to the second-order GR term, it is 1/15; relative to the leading term in the total deflection, it is only $\pi\epsilon/16$. Thus the first-order deflections can be approximately the same while the second-order terms remain distinguishable. Comparison at the same mass and the same impact parameter is an essential part of this criterion.

This test mainly compares how the common time and length scales extend beyond the weak field. Actual measurements also require light-path calculations at finite emission and reception distances, together with treatment of source rotation, source shape, and the intervening medium. Equation (S17) provides a directly comparable geometric benchmark for a spherically symmetric source with distant incidence and reception.

§S3.3 Strong-Field Light Paths

Near a compact source, retaining only the first two orders is no longer adequate for light paths. A turning point is determined by $u' = 0$, which can be written as

$$b = \mathcal{R}_{\rm opt}(r_0), \qquad \mathcal{R}_{\rm opt}(r) = r\frac{B(r)}{A(r)} = re^{2\mu/r}. \tag{S18}$$

Here $\mathcal{R}_{\rm opt}$ is the optical radius. For a ray arriving from a large distance, its minimum gives the critical impact parameter; precisely at this minimum, radial turning coincides with a circular light orbit. Differentiation gives

$$\frac{d\mathcal{R}_{\rm opt}}{dr} = e^{2\mu/r}\left(1 - \frac{2\mu}{r}\right), \qquad \boxed{r_{\rm ph} = 2\mu, \quad b_c = 2e\mu.} \tag{S19}$$

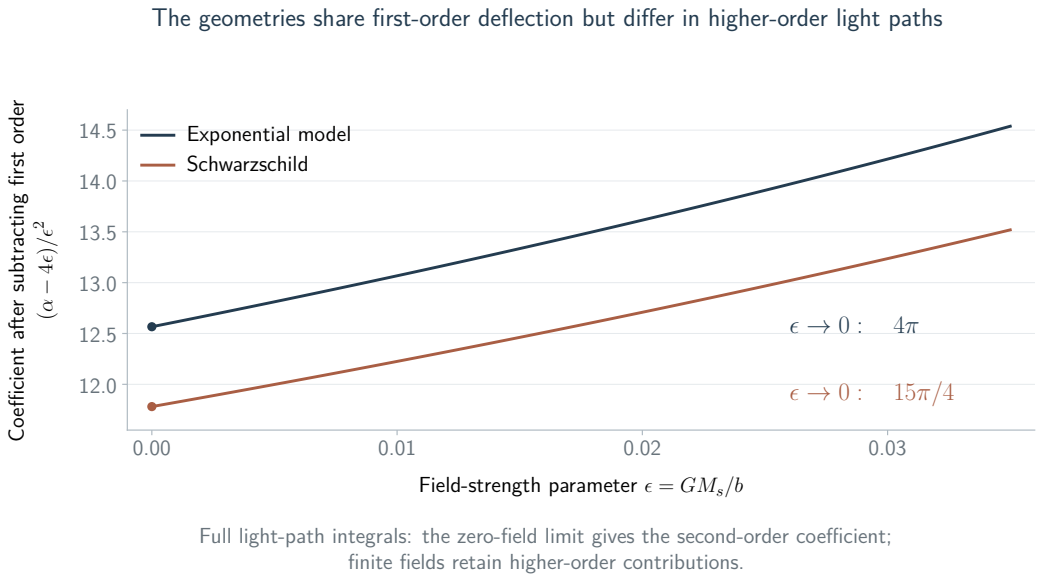


Figure S2. Difference in second-order light deflection. The vertical axis subtracts the common first-order term from the full deflection angle and divides the result by ϵ^2. The curves are calculated by integrating the light orbits in Eq. (S14); as $\epsilon \to 0$, the exponential model approaches 4π, and the Schwarzschild model approaches $15\pi/4$. At finite ϵ, third- and higher-order terms remain, so the variation of the curves cannot be attributed entirely to the second-order coefficient.

This requires the external exponential geometry to cover the region $r = 2\mu$; the interior structure of an actual celestial body is determined separately by the matter solution. This circular orbit is unstable: the radial effective potential for light is proportional to $1/\mathcal{R}^2_{\rm opt}$ and has a maximum here. Rays near the critical value can travel through large angles around the source before returning outward or moving farther inward.

Comparison with GR must use the same kind of length scale. In the exponential model, the areal radius of a sphere is $R = rB = re^{\mu/r}$, so the areal radius of the circular light orbit is $2\sqrt{e}\,\mu$. The Schwarzschild light orbit has areal radius 3μ and critical impact parameter $3\sqrt{3}\,\mu$. The distant angular scale is directly controlled by the respective b_c, whose ratio is

$$\frac{b_{c,\rm exp}}{b_{c,\rm GR}} = \frac{2e}{3\sqrt{3}} \simeq 1.04627. \tag{S20}$$

If light entering inside the critical orbit is absorbed or cannot return to the observer, and background illumination makes the dark region visible, the distant angular radius of the critical dark region is approximately b_c/D, where D is the observation distance in the asymptotically flat region. At the same mass and distance, Eq. (S20) gives a difference of approximately 4.63% in geometric scale. The actual contours of the visible bright ring and dark region also depend on the emitting matter and radiative transport and cannot be determined by a single critical parameter alone.

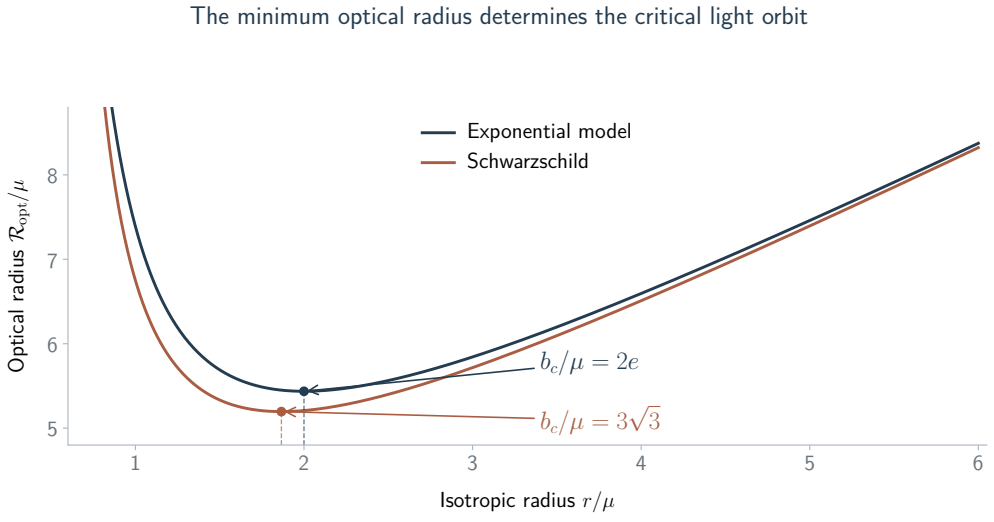


Figure S3. The minimum of the optical radius determines the critical light path. The horizontal axis uses the asymptotically normalized isotropic radius of each geometry, and the vertical axis is $\mathcal{R}_{\rm opt}/\mu$. The minima give the critical impact parameters $2e$ and $3\sqrt{3}$, respectively.

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